

Payment Management System

Project Title: Helios Watch Store E-Commerce Database

Skill Task: DBMS

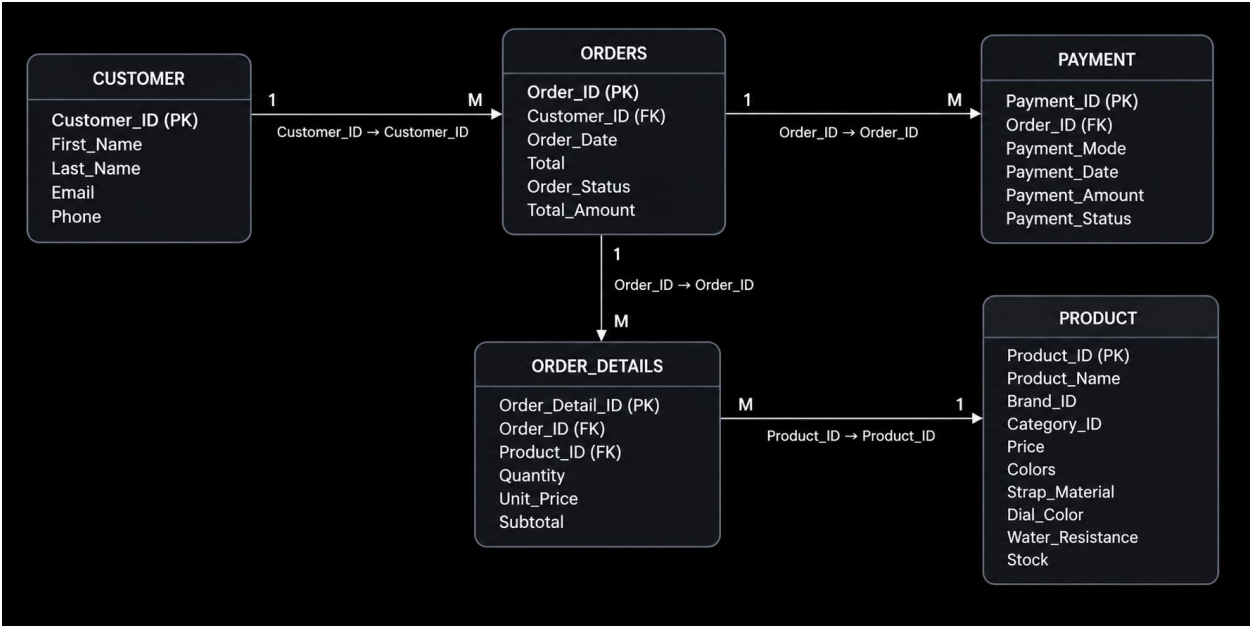
Week: 5

1. Database Design

1.1 Payment Table

Attribute	Data Type	Key Type / Constraint
Payment_ID	NUMBER	Primary Key
Order_ID	NUMBER	Foreign Key
Payment_Mode	VARCHAR2(20)	NOT NULL
Payment_Date	DATE	NOT NULL
Payment_Amount	NUMBER(10,2)	NOT NULL
Payment_Status	VARCHAR2(20)	NOT NULL

2. Table Relationships



3. Table Creation

3.1 Create Payment Table

```
SQL> CREATE TABLE Payment (  
2     Payment_ID      NUMBER PRIMARY KEY,  
3     Order_ID        NUMBER NOT NULL,  
4     Payment_Mode     VARCHAR2(20) NOT NULL,  
5     Payment_Date     DATE NOT NULL,  
6     Payment_Amount   NUMBER(10,2) NOT NULL,  
7     Payment_Status   VARCHAR2(20) NOT NULL,  
8     CONSTRAINT FK_Payment_Order  
9         FOREIGN KEY (Order_ID)  
10        REFERENCES Orders(Order_ID)  
11 );
```

Table created.

4. Payment Data Insertion

```
SQL> INSERT INTO Payment  
2 VALUES (501, 1001, 'UPI', TO_DATE('01-09-2026','DD-MM-YYYY'), 6497.00, 'Successful');  
  
1 row created.  
  
SQL>  
SQL> INSERT INTO Payment  
2 VALUES (502, 1002, 'Credit Card', TO_DATE('02-09-2026','DD-MM-YYYY'), 3999.00, 'Successful');  
  
1 row created.  
  
SQL>  
SQL> INSERT INTO Payment  
2 VALUES (503, 1003, 'Debit Card', TO_DATE('03-09-2026','DD-MM-YYYY'), 1999.00, 'Failed');  
  
1 row created.  
  
SQL>  
SQL> INSERT INTO Payment  
2 VALUES (504, 1004, 'UPI', TO_DATE('04-09-2026','DD-MM-YYYY'), 5999.00, 'Successful');  
  
1 row created.  
  
SQL>  
SQL> INSERT INTO Payment  
2 VALUES (505, 1005, 'Net Banking', TO_DATE('05-09-2026','DD-MM-YYYY'), 7499.00, 'Successful');  
  
1 row created.
```

Activate Windows

```

SQL>
SQL> INSERT INTO Payment
  2 VALUES (506, 1006, 'UPI', TO_DATE('06-09-2026','DD-MM-YYYY'), 4599.00, 'Failed');

1 row created.

SQL>
SQL> INSERT INTO Payment
  2 VALUES (507, 1007, 'Credit Card', TO_DATE('07-09-2026','DD-MM-YYYY'), 5299.00, 'Successful');

1 row created.

SQL>
SQL> INSERT INTO Payment
  2 VALUES (508, 1008, 'Debit Card', TO_DATE('08-09-2026','DD-MM-YYYY'), 12999.00, 'Successful');

1 row created.

SQL>
SQL> INSERT INTO Payment
  2 VALUES (509, 1009, 'UPI', TO_DATE('09-09-2026','DD-MM-YYYY'), 2499.00, 'Successful');

1 row created.

```

5. Manage Payment Transactions

5.1 Display Successful Payments

```

SQL> SELECT *
  2 FROM Payment
  3 WHERE Payment_Status = 'Successful';

```

PAYMENT_ID	ORDER_ID	PAYMENT_MODE	PAYMENT_D	PAYMENT_AMOUNT
501	1001	UPI	01-SEP-26	6497
502	1002	Credit Card	02-SEP-26	3999
504	1004	UPI	04-SEP-26	5999

PAYMENT_ID	ORDER_ID	PAYMENT_MODE	PAYMENT_D	PAYMENT_AMOUNT
505	1005	Net Banking	05-SEP-26	7499
Successful				
507	1007	Credit Card	07-SEP-26	5299
Successful				
508	1008	Debit Card	08-SEP-26	12999
Successful				
509	1009	UPI	09-SEP-26	2499
Successful				

7 rows selected.

5.2 Display Failed Payments

```
SQL> SELECT *
2 FROM Payment
3 WHERE Payment_Status = 'Failed';
```

PAYMENT_ID	ORDER_ID	PAYMENT_MODE	PAYMENT_D	PAYMENT_AMOUNT
503	1003	Debit Card	03-SEP-26	1999
Failed				
506	1006	UPI	06-SEP-26	4599
Failed				

6. Modify Payment Status

```

SQL> UPDATE Payment
  2  SET Payment_Status = 'Successful'
  3  WHERE Payment_ID = 503;

1 row updated.

SQL> SELECT *
  2  FROM Payment
  3  WHERE Payment_ID = 503;

```

PAYMENT_ID	ORDER_ID	PAYMENT_MODE	PAYMENT_D	PAYMENT_AMOUNT
503	1003	Debit Card	03-SEP-26	1999
Successful				

7. Analyze Payment Methods

```

SQL> SELECT
  2     Payment_Mode,
  3     COUNT(Payment_ID) AS Total_Transactions,
  4     SUM(Payment_Amount) AS Total_Amount
  5  FROM Payment
  6  WHERE Payment_Status = 'Successful'
  7  GROUP BY Payment_Mode
  8  ORDER BY Payment_Mode;

```

PAYMENT_MODE	TOTAL_TRANSACTIONS	TOTAL_AMOUNT
Credit Card	2	9298
Debit Card	2	14998
Net Banking	1	7499
UPI	3	14995

8. Payment Transaction Report

```

SQL> SELECT
  2     p.Payment_ID,
  3     o.Order_ID,
  4     c.Customer_ID,
  5     c.First_Name,
  6     c.Last_Name,
  7     p.Payment_Mode,
  8     p.Payment_Date,
  9     p.Payment_Amount,
10     p.Payment_Status
11 FROM Payment p
12 JOIN Orders o
13     ON p.Order_ID = o.Order_ID
14 JOIN Customer c
15     ON o.Customer_ID = c.Customer_ID
16 ORDER BY p.Payment_ID;

```

PAYMENT_ID	ORDER_ID	CUSTOMER_ID		

FIRST_NAME				

LAST_NAME			PAYMENT_MODE	

PAYMENT_ID	PAYMENT_AMOUNT	PAYMENT_STATUS		

501	1001	1		
Renu			UPI	
Srinivasan				
01-SEP-26	6497	Successful		
PAYMENT_ID	ORDER_ID	CUSTOMER_ID		

FIRST_NAME				

LAST_NAME			PAYMENT_MODE	

PAYMENT_ID	PAYMENT_AMOUNT	PAYMENT_STATUS		

502	1002	2		
Arjun			Credit Card	
Kumar				
02-SEP-26	3999	Successful		

```
PAYMENT_ID  ORDER_ID  CUSTOMER_ID
-----
FIRST_NAME
-----
LAST_NAME                                     PAYMENT_MODE
-----
PAYMENT_D  PAYMENT_AMOUNT  PAYMENT_STATUS
-----
          503          1003          3
Priya
Kumar
03-SEP-26          1999 Successful
Debit Card
```

```
PAYMENT_ID  ORDER_ID  CUSTOMER_ID
-----
FIRST_NAME
-----
LAST_NAME                                     PAYMENT_MODE
-----
PAYMENT_D  PAYMENT_AMOUNT  PAYMENT_STATUS
-----
          504          1004          4
Rahul
Sharma
04-SEP-26          5999 Successful
UPI
```

```
PAYMENT_ID  ORDER_ID  CUSTOMER_ID
-----
FIRST_NAME
-----
LAST_NAME                                     PAYMENT_MODE
-----
PAYMENT_D  PAYMENT_AMOUNT  PAYMENT_STATUS
-----
          505          1005          5
Keerthana
Mohan
05-SEP-26          7499 Successful
Net Banking
```

```
PAYMENT_ID  ORDER_ID  CUSTOMER_ID
-----
FIRST_NAME
-----
LAST_NAME                                     PAYMENT_MODE
-----
PAYMENT_D  PAYMENT_AMOUNT  PAYMENT_STATUS
-----
          506          1006          6
Nandhini
Ravi
06-SEP-26          4599 Failed
UPI
```

PAYMENT_ID	ORDER_ID	CUSTOMER_ID	

FIRST_NAME			

LAST_NAME			PAYMENT_MODE

PAYMENT_D	PAYMENT_AMOUNT	PAYMENT_STATUS	

507	1007	7	
Karthik			
Prakash			Credit Card
07-SEP-26	5299	Successful	

PAYMENT_ID	ORDER_ID	CUSTOMER_ID	

FIRST_NAME			

LAST_NAME			PAYMENT_MODE

PAYMENT_D	PAYMENT_AMOUNT	PAYMENT_STATUS	

508	1008	8	
Swetha			
Balaji			Debit Card
08-SEP-26	12999	Successful	

PAYMENT_ID	ORDER_ID	CUSTOMER_ID	

FIRST_NAME			

LAST_NAME			PAYMENT_MODE

PAYMENT_D	PAYMENT_AMOUNT	PAYMENT_STATUS	

509	1009	9	
Vignesh			
Ganesh			UPI
09-SEP-26	2499	Successful	

9 rows selected.